

Manuscript #3 - first draft of results (23.11.2025)

“Top-down vs bottom-up at CARIACO -

deciphering the dynamics of a tropical coastal marine ecosystem.”

Abstract: This study investigates the mechanisms driving a phytoplankton community size structure shift in the Cariaco Basin between 1996-2017. Following 2004, a shift from a large-dominated community to one dominated by smaller sizes was observed, and is clearly apparent in size data calculated from pigment composition. Coinciding the size shift, there was a collapse in the sardine fisheries.

Using long-term observational data and a size-spectral ecosystem model (adapted from Stock et al. 2008), we test whether the shift from large to small phytoplankton dominance resulted from bottom-up forcing (reduced upwelling and nutrient supply), top-down control (reduced zooplankton mortality following fisheries collapse), or changes in zooplankton community composition (signified by the gross grazing efficiency).

We run the model at steady state and used parameter scans to match the relative size spectral composition of clusters of the time series data, and to explore the mechanisms underlying this shift.

Our factorial modeling approach reveals that the size shift can best be recreated in our model driven by the combination of reduced zooplankton higher-order mortality and altered zooplankton grazing efficiency, rather than nutrient limitation alone.

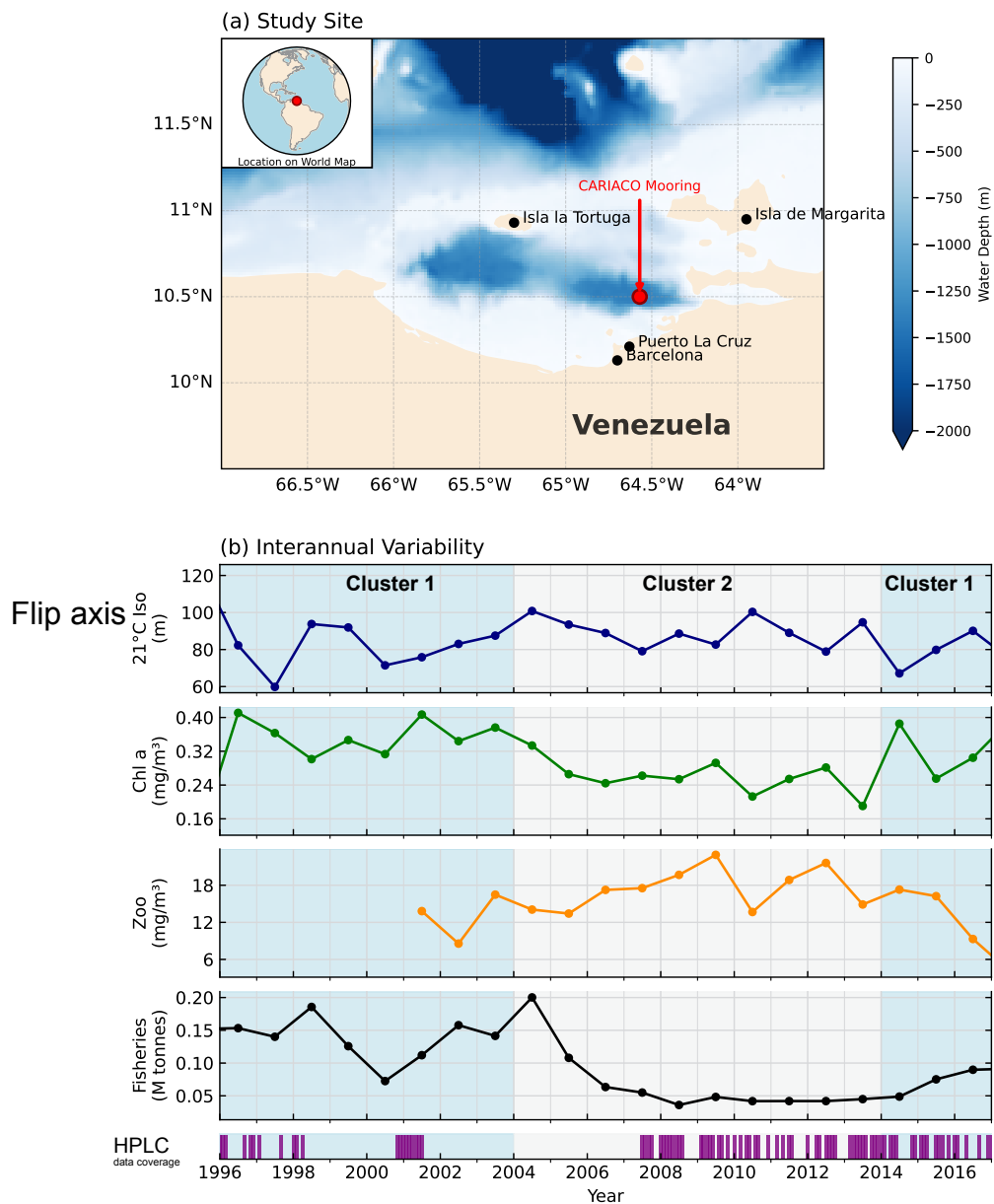
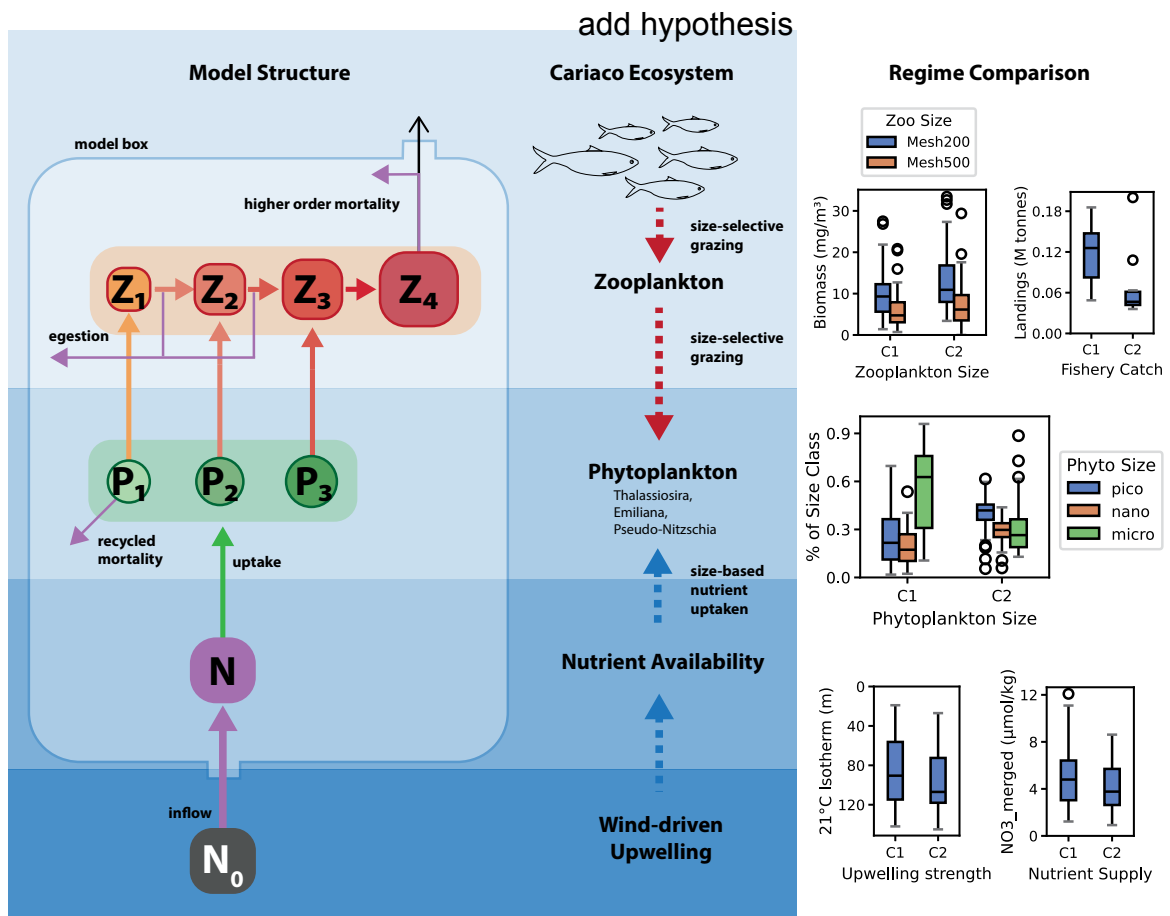


Figure 1: Map + Timeseries Overview.

Panel b): Shows interannual variability via the geometric mean, the two clusters are highlighted with a background overlay. The bottom row shows the data coverage of the HPLC data, used to calculate the phytoplankton size spectrum.

remove data, combine with model results



continuous shading

Figure 2: Model Structure + Conceptual overview + Regime Change
In Regime Comparison: C1 = Cluster 1, C2 = Cluster 2

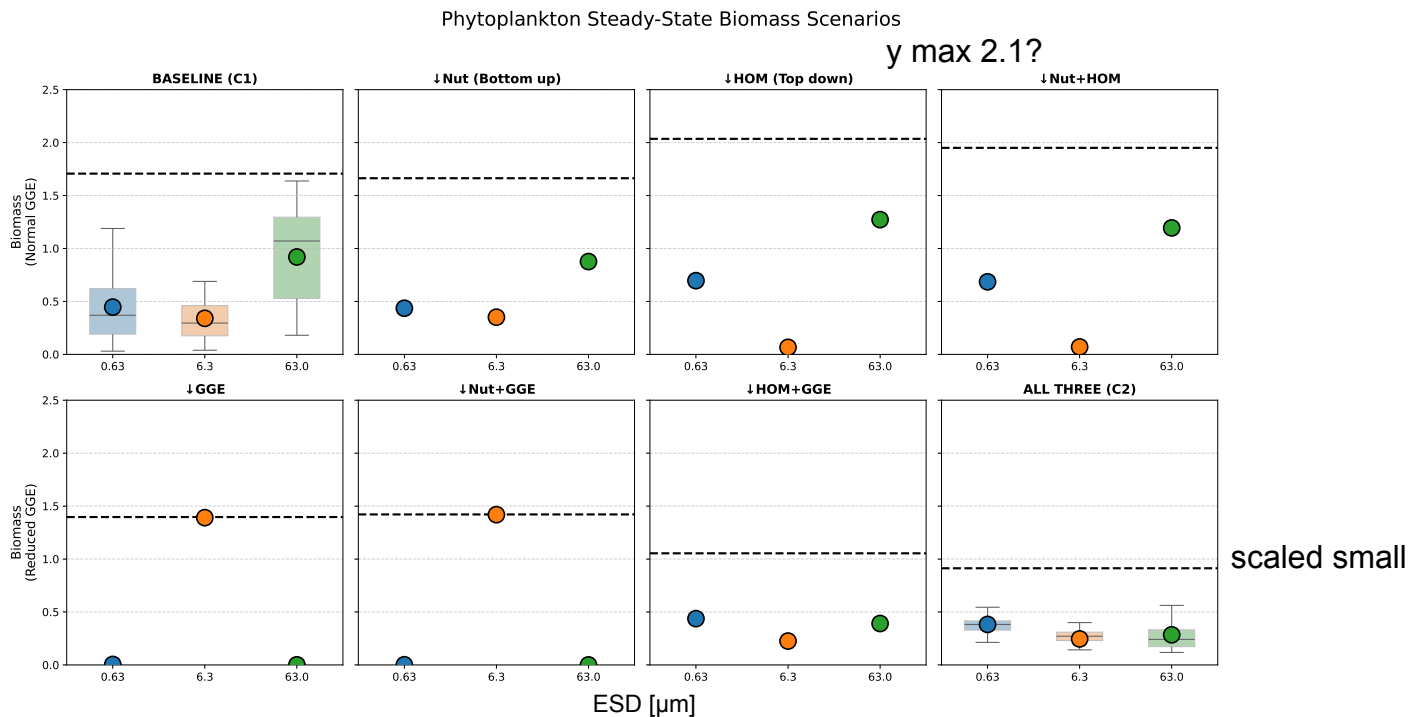


Figure 3: Scenario Comparison

Varying the parameters sequentially and in all combinations, between model fit for cluster 1 (Baseline) and cluster 2 (All three). These two plots also show the size spectral data (relative size distribution) from HPLC data.

Nut = Nutrient supply (external constant forcing)

HOM = Higher Order Mortality (acting on Z_4)

GGE = Gross Growth Efficiency (of Zooplankton).

Top Row has high GGE

Bottom Row has low GGE

Values:

NUT_BASE = 0.0053 NUT_LOW = 0.00325 (calculated from average values per Cluster, units are $\text{mmol N m}^{-3} \text{ day}^{-1}$)

HOM_BASE = 0.026118 HOM_LOW = 0.011204 (result of parameter scan)

GGE_BASE = 0.370505 GGE_LOW = 0.310345 (result of parameter scan)

all other parameters are kept the same as in Stock et. al 2008.

Phytoplankton Size Class Dominance Across Parameter Spaces

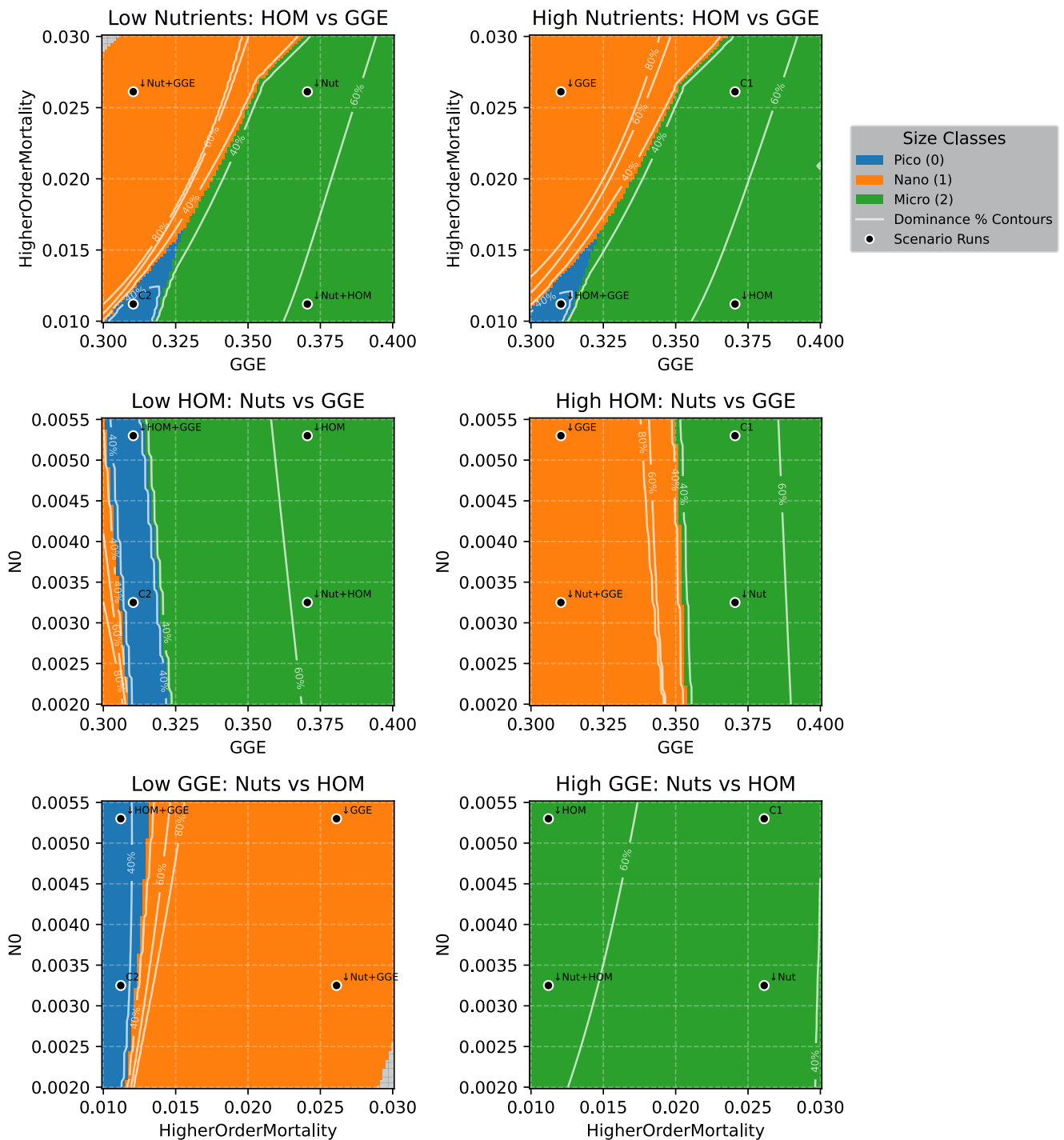


Figure 4: Exploring parameter space, large dominated to small dominated shift.

3-way parameter scans, visualized via the color of the dominating phytoplankton size class. Obviously GGE is the defining parameter for shift, and a combination of HOM and GGE is necessary. Nutrient supply has a negligible effect.